

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10: PROPERTIES OF WAVES (SUB-STRAND 2.1) FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Today you will write your Final Explanation of the matatu phenomenon. Think about Baraka's matatu on the Nairobi-Ngong route, and about Wanjiru and Otieno standing around the corner of a building on a busy Nairobi street. Baraka keeps his horn at 400 Hz and his radio on one 100 MHz FM station. Use the wave equation $v = f\lambda$ and the figures from our lessons: sound in air travels at 340 m/s, light in air travels at 3.0×10^8 m/s, the street opening is 4.0 m wide, and the matatu moves at a constant 15 m/s. Write your answer in the space under each prompt. For every explanation, give a claim, the evidence (with numbers and units) and your reasoning. Show your working for every calculation. Use the words pitch for frequency and loudness for amplitude. Refer back to the Driving Question Board and your class model (Parts A, B and C) where it helps.

Part A: Why the music turns the corner but the torch beam does not

Wanjiru and Otieno cannot see the matatu behind the concrete wall, but they hear its music clearly. A torch beam from the matatu would be blocked. (a) Use $v = f\lambda$ with 340 m/s to calculate the wavelength of the horn (400 Hz) and the wavelengths of the lowest (200 Hz) and highest (1000 Hz) notes of the music. (b) Calculate the ratio of the horn wavelength to the 4.0 m opening, and the approximate ratio for visible light of wavelength 5.0×10^{-7} m. (c) Write a claim-evidence-reasoning explanation of why the music is heard around the corner and the torch beam is not. (d) Which would be heard better around the corner, a 200 Hz note or a 1000 Hz note? Explain.

(a) Horn: $\lambda = v/f = 340/400 = 0.85$ m. Lowest music note: $340/200 = 1.7$ m. Highest music note: $340/1000 = 0.34$ m. So the music has wavelengths from 0.34 m to 1.7 m.

(b) Horn ratio = $0.85/4.0 = 0.21$. Light ratio = $5.0 \times 10^{-7}/4.0 = \text{about } 1.3 \times 10^{-7}$.

(c) Claim: The music is heard around the corner because sound waves diffract (bend) around the 4.0 m opening, but light hardly diffracts at all, so the torch beam is blocked. Evidence: sound wavelengths of 0.34 m to 1.7 m (the horn is 0.85 m, ratio 0.21) are comparable to the 4.0 m opening, while light of wavelength 5.0×10^{-7} m has a ratio of about 1.3×10^{-7} . In the ripple tank, narrow gaps made waves spread out, and in the classroom corner demonstration the tone was heard but the torch beam did not reach us. Reasoning: diffraction is the spreading of waves around an obstacle or through a gap, and it is greatest when the wavelength is comparable to or bigger than the gap. Sound meets this condition so it spreads round the corner. Light's wavelength is far smaller than the gap, so it travels almost in a straight line (rectilinear propagation) and the wall casts a shadow.

(d) The 200 Hz note ($\lambda = 1.7$ m) would be heard better. Its longer wavelength is closer to the 4.0 m opening than the 0.34 m wavelength of the 1000 Hz note, so it diffracts more. Higher pitches bend less than low pitches.

Part B: Why the horn changes pitch

Baraka drives past at a constant 15 m/s and never changes his horn, which has a true frequency of 400 Hz. (a) Use $f' = f v / (v - v_s)$ and $f' = f v / (v + v_s)$ with $v = 340$ m/s to calculate the frequency Otieno hears as the matatu approaches and the frequency Wanjiru hears as it moves away. Round to the nearest Hz. (b) Find the drop in pitch. (c) Explain, using wavefronts, why the pitch heard changes although the horn stays the same. (d) Which fixed quantity stays the same at the horn itself?

- (a) Approaching: $f' = 400 \times 340 / (340 - 15) = 400 \times 340 / 325 = 418$ Hz. Receding: $f' = 400 \times 340 / (340 + 15) = 400 \times 340 / 355 = 383$ Hz.
 (b) Drop = $418 - 383 = 35$ Hz.
 (c) This is the Doppler effect, an apparent change in frequency caused by relative motion between source and observer. The horn keeps producing 400 waves every second. As the matatu moves forward, each wavefront is sent out from a point closer to the listener ahead, so the wavefronts are compressed, the wavelength is shorter and more waves arrive each second, giving a higher pitch (418 Hz). Behind the matatu, each wavefront is sent from a point further away, so the waves are stretched, the wavelength is longer and fewer arrive each second, giving a lower pitch (383 Hz). The speed of sound stays 340 m/s, and $v = f\lambda$ links the changed wavelength to the changed frequency heard.
 (d) The horn's own frequency stays at 400 Hz; Baraka never changes it. Only the frequency received by the listener changes.

Part C: Why the FM radio fades near the Ngong Hills

The matatu radio is tuned to one 100 MHz FM station that never changes, yet it fades and crackles at certain spots near the Ngong Hills and comes back a few metres later. (a) Use $v = f\lambda$ with 3.0×10^8 m/s to find the wavelength of the signal. (b) Explain, using the words superposition, path difference and interference, why the signal fades at some spots and is strong between them. (c) How far apart are the weak spots, and how does this follow from your answer to (a)? (d) In the classroom, two speakers both played 340 Hz. Give the wavelength and say whether a path difference of 1.00 m or 0.50 m gave the quiet spot. How does this support your explanation in (b)?

- (a) $\lambda = v/f = 3.0 \times 10^8 / 1.0 \times 10^8 = 3.0$ m.
 (b) Near the Ngong Hills the radio receives two signals: one directly from the transmitter and one reflected from the hillside. The two waves overlap and add together (superposition). Where the path difference is a whole number of wavelengths ($n\lambda$) the waves reinforce, which is constructive interference, and the signal is strong. Where the path difference is $(n + 1/2)\lambda$ the waves cancel, which is destructive interference, and the signal is weak and crackles. Moving the matatu a short distance changes the path difference, so the radio fades and then returns even though the station never changes.
 (c) Weak spots are half a wavelength apart, so $3.0/2 = 1.5$ m apart.
 (d) For 340 Hz, $\lambda = 340/340 = 1.00$ m. A path difference of 0.50 m (half a wavelength) gave the quiet spot, and 1.00 m gave a loud spot. The quiet spots disappeared when one speaker was covered because there was then no second wave to cancel the first. This shows that the same interference idea explains the fading FM signal, only with a wavelength of 3.0 m instead of 1.00 m.

Part D: Using wave ideas in technology

Wave properties are applied in everyday life. (a) Fishers on Lake Victoria use sonar. A pulse returns after 0.040 s and sound travels at 1500 m/s in water. Calculate the depth and explain why you divide by 2. (b) A doctor uses ultrasound of frequency 3.5 MHz, which travels at 1540 m/s in soft tissue. Calculate its wavelength in millimetres and state how the image is formed. (c) In FM radio, a 1 kHz tone is sent on a 100 MHz carrier. State what changes in the carrier and what stays constant.

- (a) $d = vt/2 = 1500 \times 0.040/2 = 30$ m. The pulse travels down to the lake bed and back up to the boat, so the total distance covered is twice the depth; dividing by 2 gives the depth.
- (b) $\lambda = v/f = 1540/3.5 \times 10^6 = 4.4 \times 10^{-4}$ m = 0.44 mm. The ultrasound pulses reflect from boundaries between different tissues, and the echoes returning (timed using $d = vt/2$) are used to build an image without cutting.
- (c) In FM, the frequency of the 100 MHz carrier changes in step with the 1 kHz tone (by up to 75 kHz), while the amplitude stays constant.

Part E: Putting it all together

Write a complete explanation for the Driving Question: How can we hear a matatu around a corner that we cannot see, and why does its pitch change and its radio signal fade as it moves? Your answer must have three paragraphs matching Parts A, B and C of the class model. In each paragraph give a claim, evidence with numbers from this assessment, and reasoning. Then say how your final model differs from your first model in Lesson 1.

Part A (corner): The music is heard around the building because sound diffracts. The sound wavelengths of 0.34 m to 1.7 m (the horn is 0.85 m, ratio 0.21 to the 4.0 m opening) are comparable to the opening, and diffraction is greatest when the wavelength is comparable to or bigger than the gap. Light of wavelength 5.0×10^{-7} m has a ratio of about 1.3×10^{-7} , so it hardly bends and the torch beam is blocked by the wall.

Part B (pitch): The horn really stays at 400 Hz. Because the matatu moves at 15 m/s, it compresses the waves ahead of it, so Otieno hears 418 Hz, and stretches them behind it, so Wanjiru hears 383 Hz, a drop of 35 Hz. This is the Doppler effect, caused by relative motion between source and listener.

Part C (fading): The 100 MHz FM signal has wavelength 3.0 m. Near the Ngong Hills the direct signal and the signal reflected from the hillside interfere. They cancel (destructive interference, path difference $(n + 1/2)\lambda$) at weak spots about 1.5 m apart and reinforce between them, so the radio fades and returns although the station never changes.

Change from the first model: In Lesson 1 I only had guesses and knew a wave is a disturbance that carries energy. My final model gives three separate causes (diffraction, the Doppler effect and interference), each supported by evidence and calculation using $v = f\lambda$.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining diffraction at the corner (Part A)	Correctly calculates all wavelengths and ratios (0.85 m, 1.7 m, 0.34 m, 0.21, about	Calculates most wavelengths and ratios correctly and explains that sound bends	States that sound bends and light does not but gives little or no use of

	1.3x10 ⁻⁷) and gives a full claim-evidence-reasoning explanation linking wavelength to gap size, including why low pitches bend more than high pitches.	and light does not because of wavelength compared with the opening, with small gaps in reasoning.	wavelength compared with the 4.0 m gap, or has calculation errors.
Explaining the Doppler effect (Part B)	Uses both Doppler formulas correctly to get 418 Hz, 383 Hz and a 35 Hz drop, and clearly explains compressed and stretched wavefronts while the horn stays at 400 Hz.	Gets the correct frequencies and gives a mostly correct explanation of higher and lower pitch with some vagueness about wavefronts.	Gives incorrect values or says the horn itself changes, or does not link pitch change to motion.
Explaining interference and fading (Part C)	Finds $\lambda = 3.0$ m, uses path difference ($n\lambda$ and $(n+1/2)\lambda$) correctly, links to the 1.5 m spacing, and connects it to the 340 Hz speaker results.	Finds the wavelength and explains destructive and constructive interference of direct and reflected signals, with minor errors in path difference or spacing.	Mentions weak signal without explaining the overlap of direct and reflected waves, or gives an incorrect wavelength.
Calculations and application to technology (Part D)	Correctly calculates 30 m, 0.44 mm and describes FM modulation, with units and explanations of $d = vt/2$.	Gets most calculations correct with units and a basic description of reflection and modulation.	Calculation errors, missing units, or does not explain the division by 2.
Scientific communication and synthesis	Uses claim, evidence and reasoning in every part, uses correct terms and units, uses Kenyan context, and compares the final model with the Lesson 1 model.	Mostly uses claim, evidence and reasoning with correct terms and units, with some weak links between parts.	Answers are incomplete or lack evidence and reasoning, with frequent errors in terms or units.

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